Introduction to Web Technologies - Assignment 3

Bootstrap: responsive design, grid layout system, containers, typography, buttons, utility classes

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Step 0. This is a continuation

You now rebuild the layout of the same site with Bootstrap. **Same team, same theme, same repository - the theme is locked until the end of the course** and you do not create new pages. Your HTML stays as it is: you may add classes, and wrap existing content in the container, row and column elements the grid needs, but you do not replace your structure and you do not start a page over. The site still runs from local files; no domain, no hosting.

The rule for this assignment: Bootstrap builds, your CSS corrects

Bootstrap must do the heavy work - the page structure, the columns, the spacing, the navigation, the buttons and the responsive behaviour all come from its classes. **Your own stylesheet becomes a small correction layer on top of it**: your brand colours, your fonts, and the few details Bootstrap does not cover. As a guide, expect to keep well under a hundred lines of your own CSS in the whole project. If a block can be built with Bootstrap classes and you built it by hand instead, it is wrong even when it looks right.

Connecting Bootstrap

Link Bootstrap from a CDN in the head of every page, and link your own stylesheet **after** it, so that your rules win where they conflict. You may also link the Bootstrap JavaScript bundle - you are not writing any JavaScript, but some components need it. Write in a comment on each page which version of Bootstrap you are using.

Containers and the grid

Every page sits inside a container. Use both container and container-fluid somewhere in the project and put a comment where you chose one over the other, saying why. Build your content with rows and columns, and make at least three different blocks respond to screen width - for example col-12 col-md-6 col-lg-4 on a set of cards, services or prices. Use the gutter and alignment classes rather than your own margins, and show that you understand nesting by putting one row inside a column at least once.

Responsive behaviour

Your pages must work at three widths: a phone around 375px, a tablet around 768px, and a normal desktop window. **Nothing may overflow sideways and no horizontal scrollbar may appear at 375px** - that is the single most common failure and it costs 10 points. Your navigation collapses into a working toggler on small screens. Use the responsive display and text-alignment utilities to hide, show or re-align at least two elements at different breakpoints.

Typography, buttons and utilities

Style your text with Bootstrap classes - display headings, lead, text alignment and colour, muted and small text - instead of writing your own font rules. Use at least four button classes, including one outline style, one size variant and one disabled state, and make sure every button is a real link or a real form control, not decoration. Across the project use at least ten different utility classes covering spacing, colour or background, borders, shadow, and flex or display - and use them on real elements, not on empty ones added to tick a box.

One Bootstrap component

Add one component from the Bootstrap documentation that genuinely suits your project - cards, accordion, carousel, alert, badge, breadcrumb, table styling, list group or modal. Copy its markup from the documentation, then adapt it to your own content, and write a comment above it naming the component and what you changed.

Cleaning up your old CSS

Bootstrap now does much of what you wrote in Assignment 2. Delete the rules it replaced - your hand-written grid, most of your spacing and alignment, your button styling - and keep a short file listing what you removed and which Bootstrap class took its place. **A stylesheet that still fights Bootstrap with its own layout rules is the main thing I will be looking for.** Your remaining CSS may set colours, fonts and small corrections, and may override a Bootstrap rule where you can justify it in a comment.

Forbidden

No downloaded themes, templates or admin dashboards, no other CSS framework, no site builders, no JavaScript of your own. Do not rebuild the grid by hand with float or with your own flex rules - that costs 10 points each time. Inline style attributes and !important cost 5 points each; if you need one, Bootstrap already has a utility class for it.

Screenshots instead of a report

There is no written report in this assignment. Submit three screenshots of one page at the three widths above, plus one screenshot of your navigation collapsed on a phone width. Nothing else is written up - everything else is checked in your code and at the defense.

Submission

The same Git repository, extended: your updated pages, your reduced stylesheet, the list of what you removed, the four screenshots, an updated AI log and README. Everyone commits from their own account - at least four commits across three different days, six if you work alone. Your HTML must still pass the W3C validator with zero errors.

Defense in class

I will resize your page in front of you and ask what happens at each breakpoint, ask what a particular class in your markup does and what would change without it, ask why you chose container over container-fluid in one place, ask which of your old rules Bootstrap replaced, and ask for a small live edit - change a three-column row into two columns on tablets, or swap a button variant. Your grade is the project score

multiplied by how well you explain your own work; if you cannot explain it, the score is zero. **Work that is submitted but not defended is not evaluated, even if it was uploaded before the deadline.** The deadline gives you the right to be graded; the defense gives you the grade.

Grading (100 points)

Containers and grid 22 · responsive behaviour at the three widths 18 · typography and buttons 12 · utility classes 12 · Bootstrap component 10 · custom CSS reduced to a correction layer 16 · screenshots and commit history 10. Penalties: 15 for a downloaded template or theme, 10 for a hand-built grid, 10 for horizontal overflow on a phone width, 5 per inline style or !important, 5 per broken link or broken component.

AI policy

You may ask AI to explain concepts and you may copy component markup from the official Bootstrap documentation, which is what the documentation is for. You may not have AI rebuild your pages, and every question you ask must be recorded in the AI log. What carries the points here is judgement - which class fits which block, what to delete, and being able to prove it at the defense.